

Multiple Choice (10 marks)

1. Suppose you live on the Moon. How long is a day (i.e. from sunrise to sunrise)?
 - (a) about 18 years
 - (b) 24 hours
 - (c) 29 Earth days
 - (d) a year
2. 20000 years from now ...
 - (a) The Moon will be closer to the Earth and the Earth's day will be longer.
 - (b) The Moon will be closer to the Earth and the Earth's day will be shorter.
 - (c) The Moon will be further from the Earth and the Earth's day will be longer.
 - (d) The Moon will be further from the Earth and the Earth's day will be shorter.
3. Approximately how old is the surface of Venus?
 - (a) 750 million years.
 - (b) 2 billion years.
 - (c) 3 billion years.
 - (d) 4.5 billion years.
4. Which of the following is/are true?
 - (a) Titan is the only outer solar system moon with a thick atmosphere
 - (b) Titan is the only outer solar system moon with evidence for recent geologic activity
 - (c) Titan's atmosphere is composed mostly of hydrocarbons
 - (d) A and D
5. You've made a scientific theory that there is an attractive force between all objects. When will your theory be proven to be correct?
 - (a) The first time you drop a bowling ball and it falls to the ground proving your hypothesis.
 - (b) After you've repeated your experiment many times.
 - (c) You can never prove your theory to be correct only "yet to be proven wrong".
 - (d) When you and many others have tested the hypothesis.

True / False (10 marks)

Answer True or False

6. True or False: Barnard's Star is the second closest star system to Earth after Proxima Centauri.
7. True or False: Water-ice is distributed evenly across Mercury's surface, not limited to the polar regions.
8. True or False: In order to observe a total solar eclipse, the Moon's umbra must intersect the location of the observer on Earth.
9. True or False: The Schwarzschild radius is primarily used to describe the characteristics of pulsars in astrophysics.
10. True or False: Pluto takes approximately 250 years to complete one orbit around the Sun.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications for solar and lunar eclipses if the Moon's orbital plane aligned precisely with the ecliptic plane, analyzing the effects on their frequency and visibility.
12. Discuss the significance of temperature measurement in astronomy, specifically comparing the Celsius and Kelvin scales, and calculate the equivalent of 25°C in Kelvin.

End of Paper